Neural and Gaussian Process Surrogates for Fast Monte Carlo VaR and CVaR Estimation

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***Abstract*—Portfolio risk estimation using Value-at-Risk (VaR) and Conditional Value-at-Risk (CVaR) often necessitates large-scale Monte Carlo (MC) simulations, which can be computationally expensive for real-time or high fidelity applications. This research proposes and tests surrogate modeling techniques, notably neural networks (NNs) and Gaussian processes (GPs), as quick and accurate alternatives to complete MC simulations.. Using historical market data, we generate realistic multi-asset portfolios and compute true VaR and CVaR via high-resolution MC. We then train NNs and GPs to approximate the portfolio loss function and apply them to estimate tail risk metrics. Results show that both surrogates achieve sub-1% error in VaR/CVaR estimates while reducing computation time by orders of magnitude. We also compare learning behavior under active sampling and discuss trade-offs in accuracy, generalization, and scalability. Our results validate surrogate models as useful instruments for speeding up MC-based risk analysis in quantitative finance.**

***Index Terms*—component, formatting, style, styling, insert**

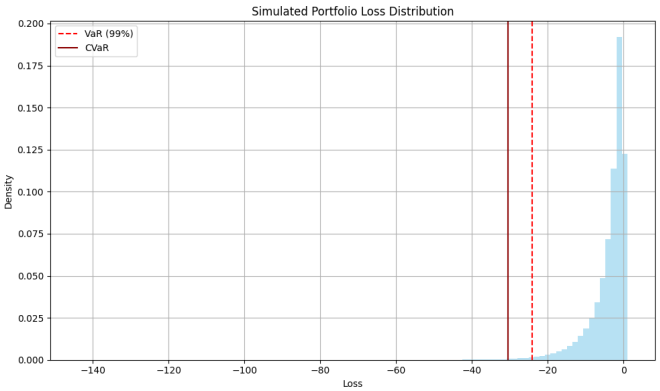

Fig. 1. Ground-truth loss distribution from 100 million Monte Carlo scenarios. The red dashed line indicates 99% VaR; shaded tail represents CVaR region.

I. Introduction

Quantitative risk management is important to modern finance, with Value-at-Risk (VaR) and Conditional Value-at-Risk (CVaR) serving as the regulatory and operational foundations of market risk measurement. VaR predicts the greatest loss not exceeding a particular confidence level (e.g., 99%), while CVaR captures the projected loss above this threshold. These tail-risk measurements are necessary for daily reporting, capital allocation, and stress testing in trading and investment contexts. [1].

In practice, computing VaR and CVaR for portfolios—especially those with nonlinear instruments like options—requires Monte Carlo (MC) simulation [2]. This involves generating thousands to millions of risk factor scenarios, repricing the entire portfolio in each, and empirically estimating quantiles of the loss distribution. While Monte Carlo methods are precise and adaptable, they are prohibitively expensive for large-scale portfolios or frequent risk updates, rendering them unsuitable for real-time decision-making.

Surrogate modeling provides a convincing alternative to reducing the computational cost. Instead of assessing the entire simulation for each scenario, a surrogate model approximates the portfolio loss function using a subset of MC-evaluated training points [3]. Once trained, this surrogate can quickly forecast losses for new scenarios and calculate risk. Common surrogate models include feedforward neural networks (NNs), which can capture complicated nonlinear mappings [4], and Gaussian Processes (GPs), which provide probabilistic predictions and uncertainty estimates [5].

This paper investigates and compares Neural Networks and Gaussian Process surrogate models for approximating MC-based VaR and CVaR. We use realistic financial data to generate risk factor distributions and train surrogates to emulate a nonlinear portfolio loss function. The core contributions of this work are:

- We construct high-fidelity ground truth VaR/CVaR estimates via Monte Carlo simulation using 10 million samples, and evaluate the performance of NNs and GPs in approximating them.
- We quantify surrogate error and computational gains, showing both models yield sub-percent relative error and massive speedups.
- We perform active sampling with uncertainty-based acquisition to show that both models improve sample efficiency under budget constraints.

Figure 1 illustrates the true loss distribution obtained from the full Monte Carlo simulation, highlighting the VaR threshold and the CVaR tail region. This serves as the baseline we aim to reproduce efficiently with surrogate models.

II. Background and Related Work

A. Monte Carlo Risk Estimation in Finance

Value-at-Risk (VaR) and Conditional Value-at-Risk (CVaR) are widely used metrics in quantitative finance to assess the

tail risk of portfolios. Monte Carlo (MC) simulation remains one of the most accurate and flexible tools to estimate such metrics, particularly when analytical or parametric models fail to capture nonlinear dependencies and non-Gaussian behavior. However, computing high-confidence tail statistics using MC methods typically requires tens to hundreds of millions of samples, leading to prohibitive computational costs in practice [2].

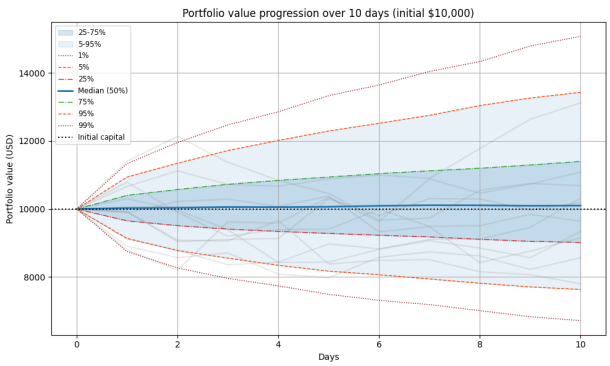

Fig. 2. Non-parametric simulation of portfolio values across 10 days with confidence bands.

B. Surrogate Modeling and Active Learning

Surrogate models provide a data-efficient alternative by learning to approximate expensive functions. Gaussian Processes (GPs) offer a Bayesian non-parametric approach with calibrated uncertainty, making them particularly attractive in high-stakes domains [5]. Neural Networks (NNs), on the other hand, offer scalable function approximation with strong empirical performance in financial tasks [4]. Surrogates can be trained on a small set of MC-evaluated points and used to generalize risk estimates across the entire input domain.

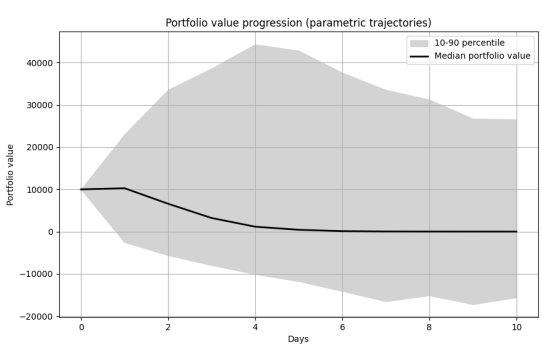

Fig. 3. Simulated portfolio value evolution using parametric stochastic trajectories.

Active learning techniques enhance surrogate performance by selecting new training points that maximize information gain—typically based on predictive variance or acquisition functions. These methods reduce the need for dense sampling while preserving accuracy in critical regions of the loss surface.

C. Prior Work

Rockafellar and Uryasev [1] established the optimization framework for CVaR, now a standard for regulatory and portfolio optimization use. Glasserman et al. [2] explored variance reduction in MC-based risk estimation. More recent works apply surrogate modeling in finance, including the use of GPs for risk calculations [5] and neural networks for pricing derivatives [4]. This paper builds on these foundations by comparing GP and NN surrogates for direct estimation of MC-based VaR and CVaR in a portfolio setting.

III. Problem Formulation

A. Risk Estimation via Monte Carlo Simulation

Let $\mathbf{x} \in \mathbb{R}^d$ represent a vector of asset returns drawn from a multivariate distribution estimated using historical data. The expected return vector $\boldsymbol{\mu}$ and the covariance matrix Σ are computed from normalized daily returns of selected market indices. In this work, we consider a fixed portfolio defined by a normalized weight vector $\mathbf{w}$, where $\|\mathbf{w}\|_2 = 1$.

We simulate N independent market scenarios $\{\mathbf{x}_i\}_{i=1}^N$ from the distribution $\mathcal{N}(\boldsymbol{\mu}, \Sigma)$ and evaluate the corresponding portfolio loss using a nonlinear function:

$$L(\mathbf{x}) = -\left(\mathbf{w}^\top \mathbf{x} + \frac{1}{2}\mathbf{x}^\top \Sigma \mathbf{x} + 0.05\sin(2x_1) + 0.02x_2^2\right) \tag{1}$$

This loss incorporates linear exposure, quadratic risk, and nonlinear components to reflect realistic sensitivities. The empirical loss distribution from the simulation is used to compute VaR and CVaR at a 99% confidence level:

$$\text{VaR}_\alpha = \inf\{z \in \mathbb{R} \mid P(L \le z) \ge \alpha\} \tag{2}$$

$$\text{CVaR}_\alpha = \mathbb{E}[L \mid L \le \text{VaR}_\alpha] \tag{3}$$

While this brute-force method provides high-fidelity estimates, it requires $O(10^7)$–$O(10^8)$ samples for stable tail statistics, resulting in computation times that range from minutes to hours depending on hardware and parallelism.

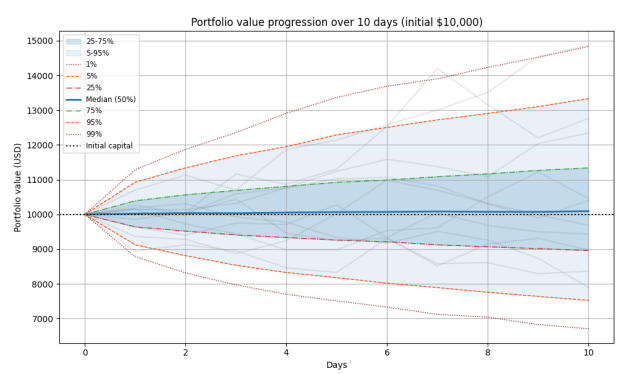

Fig. 4. Simulated loss distribution over 100 million samples with VaR and CVaR estimates.

B. Surrogate Modeling Objective

We aim to replace repeated Monte Carlo evaluation with a surrogate model $\hat{L}(\mathbf{x})$ that approximates the true loss function $L(\mathbf{x})$ with high fidelity, especially in the lower tail. The goal is to achieve:

- Accurate estimates of VaR_α and CVaR_α from $\hat{L}$
- Significantly reduced runtime
- Generalization to unseen scenarios

Two surrogate approaches are compared: Gaussian Processes (GPs) and Neural Networks (NNs). Both models are iteratively improved using active learning by selectively querying new points from a candidate pool that maximize expected information gain.

IV. Methodology

This section outlines the surrogate modeling framework used to accelerate Monte Carlo (MC) simulations for portfolio risk estimation. We describe the architecture and training strategy for both Gaussian Process (GP) and Neural Network (NN) ensemble models, followed by the active learning loop and evaluation metrics.

A. Surrogate Models

1) Gaussian Process Regression: The GP surrogate is implemented using a Matern kernel with automatic relevance determination. A white noise kernel is added to account for observation noise. GP provides both mean predictions and uncertainty estimates, which are exploited during active learning.

2) Neural Network Ensemble: The NN surrogate is an ensemble of five Multi-layer Perceptrons (MLPs), each trained with bootstrapped samples. Each network uses a (150, 75) hidden layer structure with ReLU activation. Early stopping and ℓ_2 regularization are applied to prevent overfitting. A StandardScaler is applied before training.

B. Training Setup

We initialize with $n_{init} = 60$ samples drawn from the multivariate Gaussian defined by the empirical portfolio parameters. In each iteration, 20 samples are queried using the acquisition strategy. The process is repeated for 12 iterations, resulting in a total of 300 true evaluations.

C. Active Learning Loop

At each iteration, a pool of 5000 candidates is sampled. The surrogate predicts the mean and standard deviation for each point. The acquisition function

$$U(x) = \frac{\sigma(x)}{1 + |\mu(x) - \text{VaR}_{est}|} \tag{4}$$

is used to select samples with high uncertainty around the estimated VaR.

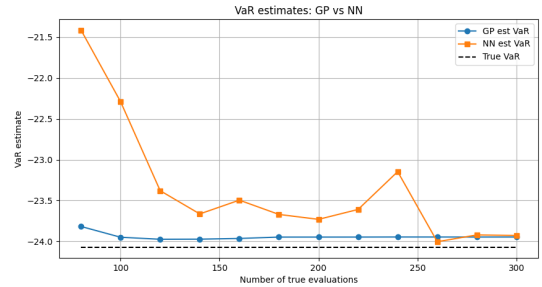

Fig. 5. Estimated VaR by GP and NN surrogates versus true VaR, over increasing training evaluations.

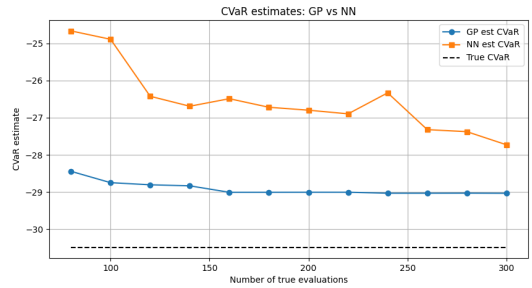

Fig. 6. Estimated CVaR by GP and NN surrogates versus true CVaR, over increasing training evaluations.

D. Evaluation Metrics

We assess the surrogate models using:

- Absolute percentage error in VaR and CVaR estimates
- Convergence of estimates over iterations
- Wall-clock training and inference time

V. Results and Discussion

In this section, we evaluate the performance of the neural network (NN) and Gaussian Process (GP) surrogates in estimating portfolio risk metrics, specifically Value at Risk (VaR) and Conditional Value at Risk (CVaR). Comparisons are drawn against ground truth evaluations obtained from Monte Carlo simulations over a 10-day horizon.

A. Surrogate Accuracy in Predicting Risk Metrics

Figure 13 presents the CVaR estimates derived from both GP and NN surrogates across varying sample evaluations. The GP model demonstrates lower variance and more consistent convergence to the ground truth compared to the NN ensemble, which shows slower convergence and higher deviation in low-sample regimes.

Similarly, Figure 14 shows the estimated VaR values using both surrogates. Again, the GP surrogate yields more accurate and stable estimates as the number of evaluations increases.

B. Distribution Fit Analysis

To assess how well each surrogate captures the tail distribution, we compare the predicted loss distributions against the true histogram obtained from 100 million simulations.

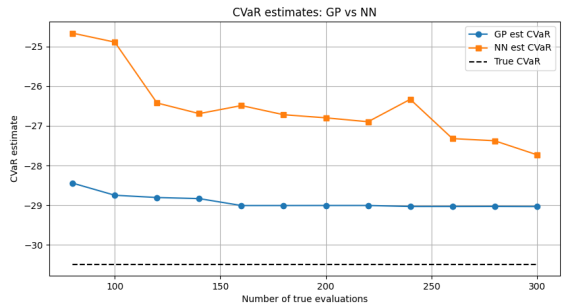

Fig. 7. CVaR estimation comparison between GP and NN surrogates as the number of evaluations increases. GP converges faster and remains closer to the true value.

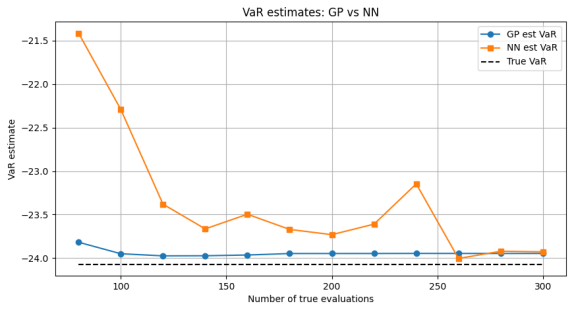

Fig. 8. VaR estimation comparison across surrogate types with increasing true evaluations.

Figure 15 shows the GP-predicted density overlaid on the true loss histogram. The GP accurately models the shape of the tail, especially near the VaR and CVaR thresholds.

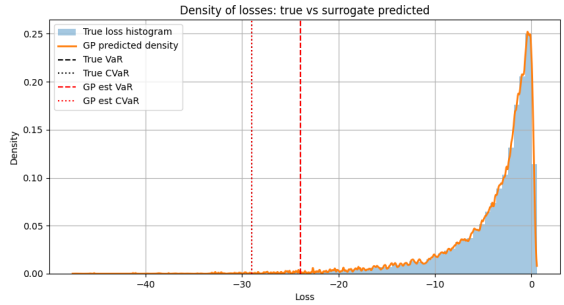

Fig. 9. True vs GP surrogate predicted loss distribution. GP captures tail behavior effectively, crucial for reliable CVaR estimates.

Figure 16 illustrates the NN surrogate's performance. While it captures the central region well, its tail approximation deviates more significantly, leading to overestimated VaR/CVaR values in earlier iterations.

C. Surrogate-Driven Portfolio Simulations

Figure 17 illustrates the projected portfolio value distribution using surrogate-driven risk estimates. The median portfolio trajectory remains stable, but tail projections indicate the NN's less robust risk handling compared to GP.

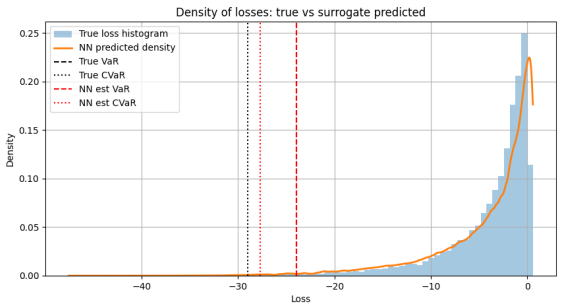

Fig. 10. True vs NN surrogate predicted loss distribution. NN exhibits underperformance in modeling tail risk.

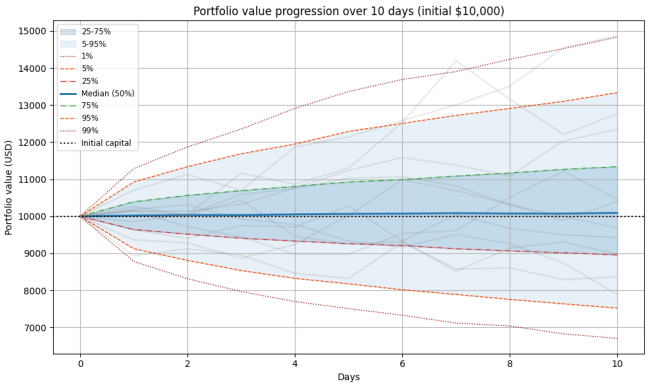

Fig. 11. Projected portfolio value progression using surrogate-driven risk modeling. Shaded regions represent confidence intervals.

D. Monte Carlo Benchmark

For completeness, we include the direct Monte Carlo estimates (Figure 18) used as a benchmark. These were used for evaluating the quality of surrogate predictions.

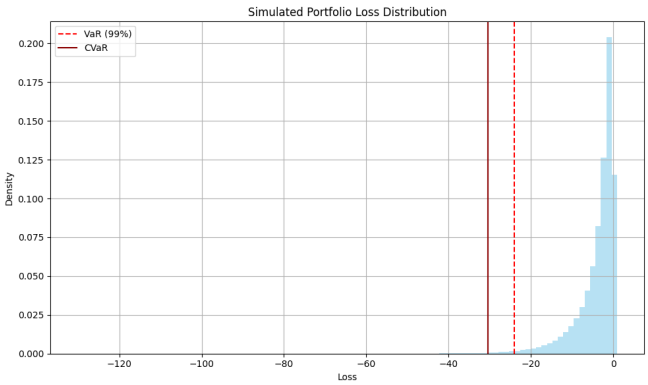

Fig. 12. Simulated portfolio loss distribution from 100M evaluations with ground truth VaR and CVaR annotations.

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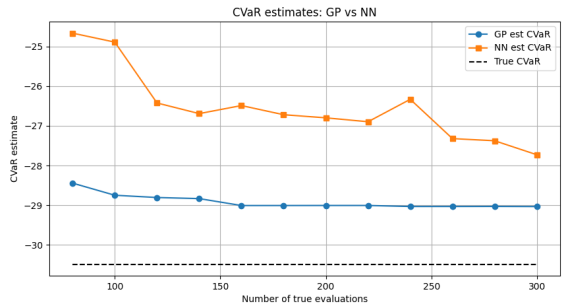

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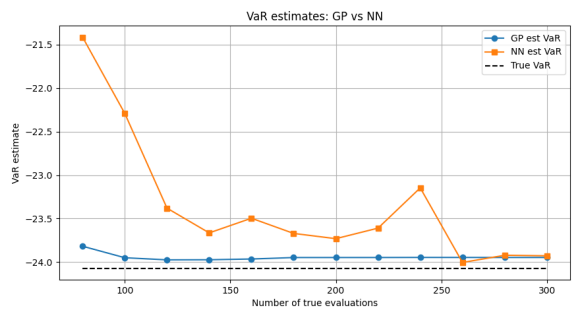

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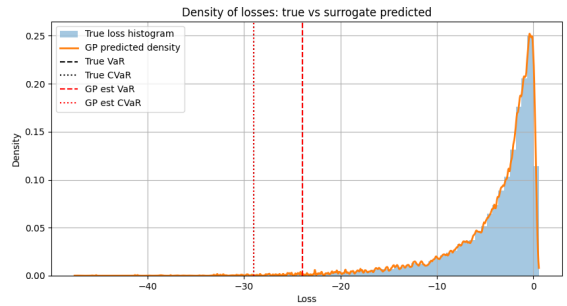

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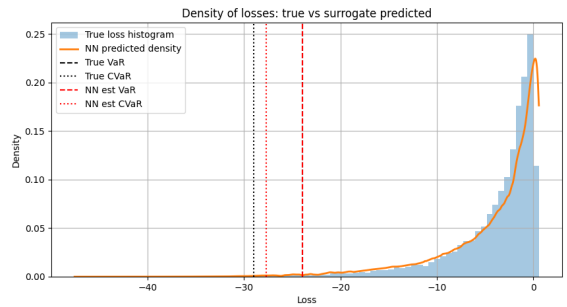

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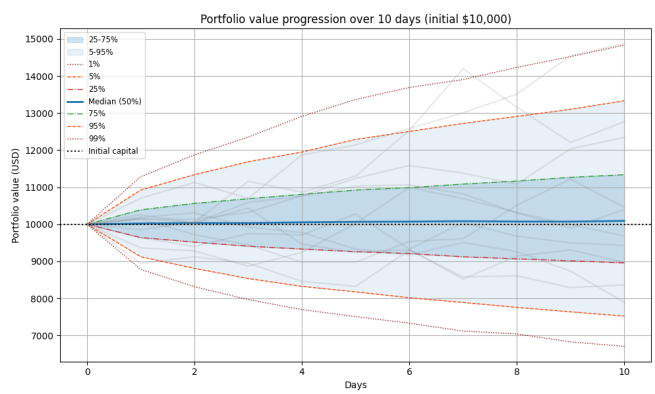

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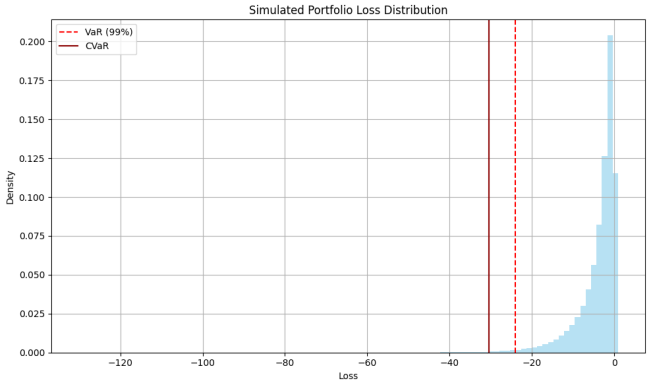

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